Claims

1. A system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints, characterized in that the system comprises:
   a **Local Security Zone**, deployed on local physical equipment, configured to perform offline slicing vectorization and feature extraction, conduct cryptographic coupling calculation on feature vectors, plaintext contents and signature salt values by using a local private key to generate globally unique unidirectionally encrypted hash fingerprints, and store original cognitive asset bodies in ciphertext only within the local equipment;
   a **Distributed Incremental Right Confirmation Ledger**, connected to the Local Security Zone, configured to receive and record incremental hash fingerprints and timestamps periodically scanned and reported by the Local Security Zone as ownership evidence of the original cognitive asset bodies, wherein the Distributed Incremental Right Confirmation Ledger does not hold any original plaintexts or original vector indexes;
   and a **Computing Power Isolation Operating Environment**, deployed in a hardware Trusted Execution Environment (TEE), connected to the Local Security Zone via a read-only data channel, provided with a session lifecycle counter and a physical degaussing erasure module. When an external session call occurs, the environment applies for a temporary physical security sandbox to decrypt and load Top-K semantically associated plaintext fragments. When a session terminates or the session lifecycle counter resets to zero, a hardware security microcontroller forcibly triggers a physical degaussing erasure interrupt to dynamically perform multi-pass random noise physical overwriting erasure on the physical memory of the sandbox loaded with private knowledge slice data, including clearing processor caches and disrupting charge metastable states of DRAM media.

2. The system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 1, characterized in that:
   a unidirectional encryption fingerprint calculation unit inside the Local Security Zone splices bytewise fixed-dimensional feature vectors v_emb generated by a local vector embedding model and corresponding sliced plaintext contents by using a signature salt value generated via a local asymmetric private key, and invokes a hash algorithm for cascading to generate a 256-bit SHA-256 hash value as the hash fingerprint.

3. The system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 1, characterized in that:

the Distributed Incremental Right Confirmation Ledger only receives incremental hash fingerprint registration through lightweight data packets containing { H_chunk, Timestamp, Owner_Signature }, where H_chunk denotes the hash fingerprint and Owner_Signature denotes the private key signature of the local device; transmission of any plaintexts and vector features is completely blocked during synchronization.

4. The system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 1, characterized in that:
the temporary physical security sandbox is a secure encrypted memory segment allocated based on hardware TEE enclave technology; the secure encrypted memory segment is mounted with read-only attributes and deprived of all network and physical medium write permissions except for inference channels of designated heterogeneous large model engines.

5. The system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 1, characterized in that:
when performing overwriting erasure, the physical degaussing erasure module invokes a hardware-level secure erasure interface to repeatedly overwrite pseudo-random noise on physical memory pages corresponding to the temporary physical security sandbox and reset physical charges to defend against cold-start residual memory reading attacks.

6. The system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 1, characterized in that:
a vector retrieval engine deployed inside the Local Security Zone is forcibly bound to the local Loopback address 127.0.0.1; an access control policy of the vector retrieval engine is set to only receive requests from user sovereign processes via local Inter-Process Communication (IPC), physically blocking direct external network and cross-namespace semantic retrieval probing against the vector database.

7. The system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 1, characterized in that the system further comprises:
a **Kernel-State Network Interception Module**, configured to monitor network sockets of external session calls and forcibly send TCP RST packets to target communication terminals when a session terminates or the lifecycle counter resets to zero, so as to erase transmission traces in network caches.

8. The system for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 1, characterized in that:
   when slicing plaintext contents offline, the Local Security Zone takes the binary bit sequence of the hash fingerprint as an input seed of a pseudo-random generator to dynamically adjust synonym replacement probabilities of preset word classes or punctuation arrangement features in sliced texts, so as to embed tiny grammatical deviation watermarks recognizable via statistical matrix inversion, serving as parameter space watermark judgment evidence against illegal crawling and training by large models.

9. A method for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints, executed by a computer processor, characterized by comprising the following steps:
   Offline Local Fingerprint Generation: the Local Security Zone slices a local plaintext knowledge base into semantic blocks and extracts feature vectors in an offline state, couples the feature vectors, plaintext contents and signature salt values to generate globally unique unidirectionally encrypted hash fingerprints, and stores original cognitive asset bodies in local ciphertext;
   Distributed Incremental Synchronization and Registration: an incremental scanning detection module reports newly generated hash fingerprints incrementally to the distributed right confirmation ledger for registration and evidence storage upon detection;
   and **Session-Level Isolation and Degaussing Erasure**: when a session call occurs, a read-only security sandbox is created in a hardware Trusted Execution Environment to decrypt and load Top-K semantically associated plaintext blocks, and a lifecycle clock is activated simultaneously; when a session terminates or the lifecycle clock resets to zero, the physical degaussing erasure module is triggered to dynamically perform multi-pass random noise physical overwriting erasure on the physical memory of the sandbox loaded with private knowledge slice data, including clearing processor caches and disrupting charge metastable states of DRAM media.

10. The method for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints according to Claim 9, characterized in that:
    in the step of offline local fingerprint generation, the generation formula of the hash fingerprint is:

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H_chunk = SHA256(v_emb || TextContent || Salt)
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where v_emb denotes a feature vector, TextContent denotes plaintext contents, and Salt denotes a signature salt value.

1. A local-end operating method in local right confirmation and synchronization of cognitive assets based on hash fingerprints, executed by a Local Security Zone device, characterized by comprising the following steps:

   Offline Local Slicing and Vectorization: slice a local plaintext knowledge base into text segments in an offline state, invoke a local vector embedding model to generate feature vectors, and store bodies of the text segments in local ciphertext;

   Local Cascaded Fingerprint Generation: splice feature vectors and plaintext contents of the text segments in a cascaded manner by using a signature salt value generated via a local private key, and calculate globally unique unidirectionally encrypted hash fingerprints;

   and **Incremental Active Synchronization and Chain Upload**: periodically scan local right confirmation logs; upon detection of newly added hash fingerprints, assemble and actively send incremental reporting requests to the distributed right confirmation ledger. The reporting requests contain hash fingerprints, timestamps and property ownership signatures generated via the local private key, and transmission of plaintext contents and feature vectors is completely blocked during chain uploading.

2. An incremental evidence storage method for a right confirmation ledger in local right confirmation and synchronization of cognitive assets based on hash fingerprints, executed by physical nodes of a distributed ledger, characterized by comprising the following steps:

   Reception of Incremental Reporting Packets: receive lightweight ownership reporting packets periodically sent by Local Security Zone devices, wherein the reporting packets only contain hash fingerprints, timestamps and ownership signatures;

   Digital Signature Verification: decrypt and verify ownership affiliation of the hash fingerprints on the basis of the ownership signatures in the reporting packets by using public keys corresponding to the Local Security Zone devices;

   and **Property Right Evidence Registration and Query**: upon successful verification, register and write the hash fingerprints and timestamps into a decentralized ledger database as unique ownership indexes without storing or writing any original plaintexts and original vector indexes, and provide read-only ownership comparison verification for recorded fingerprints to an external data sovereignty policy central gateway.

3. A computing power sandbox isolation execution method in local right confirmation and synchronization of cognitive assets based on hash fingerprints, executed by a computing power

isolation operating processor deployed in a hardware Trusted Execution Environment, characterized by comprising the following steps:

Allocation of Secure Read-Only Sandbox: upon reception of a session call request, apply to the Trusted Execution Environment for allocation of a temporary physical security sandbox, and establish a unidirectional read-only data transmission channel with a Local Security Zone device;

Decryption Loading and Clock Binding: acquire decrypted plaintexts of Top-K relevant text segments via the read-only data channel, load the plaintexts into secure encrypted memory segments corresponding to the physical security sandbox, and bind a hardware lifecycle timer to the secure encrypted memory segments;

and **Physical Degaussing and Network Circuit Break**: when session completion or timer expiration is detected, immediately perform multi-pass random noise physical overwriting erasure on the physical memory of the physical security sandbox to clear residual charges, including clearing processor caches and disrupting charge metastable states of DRAM media, and forcibly send TCP RST reset packets to the peer session terminal at the network protocol stack layer to erase data transmission traces.

4. The local-end operating method according to Claim 11, characterized in that in the step of offline local slicing and vectorization, when slicing the plaintext knowledge base, the binary bit sequence of the hash fingerprint is taken as an input seed of a pseudo-random generator to dynamically adjust synonym replacement probabilities of preset word classes or punctuation arrangement features in sliced texts, so as to embed grammatical deviation watermarks recognizable via statistical matrix inversion.

5. The incremental evidence storage method for a right confirmation ledger according to Claim 12, characterized in that:
in the step of digital signature verification, time sequence comparison is performed on timestamps contained in received reporting packets to defend against replay attacks targeting property right evidence registration; property right evidence registration adopts a decentralized blockchain cascaded block structure for physical storage.

6. The computing power sandbox isolation execution method according to Claim 13, characterized in that:
in the step of physical degaussing and network circuit break, the multi-pass random noise physical overwriting erasure comprises sequentially writing all-zero values, all-one values and random bit noise generated by a cryptographically secure pseudo-random number generator

to all physical addresses occupied by the secure encrypted memory segments; after each pass of overwriting, a processor cache line flush instruction is invoked to force data in all levels of processor caches to be flushed back to physical memory pages for charge neutralization.

7. A computer-readable storage medium storing a computer program thereon, characterized in that the program, when executed by a processor, implements the method according to any one of Claims 9 to 16.